

Dancing with What: Talent or Stardom?

Summary

Dancing with the Stars is a television franchise that combines professional judging with public voting, and it has faced questions about fairness between “skill” and “popularity.” Based on 34 seasons of data, this paper establishes mathematical models to estimate audience votes, compare voting mechanisms, and analyze contestant characteristics to design a fairer format.

For Taks1, two models are established based on format differences: a geometric constraint model for ranking-based seasons, while the percentage-based seasons employ a Bayesian probability model. The models achieved prediction accuracies of **85%**, **100%**, **100%**, and **87%** for elimination outcomes in Weeks 4 of Seasons 6, 19, 23, and 28, respectively. The standard deviation of the posterior distribution reveals varying degrees of certainty among contestants across different weeks.

For Taks2, the “reversal rate” under different season rule switches is calculated. The reversal rate reached **57.14%** when S2 switched to the percentage system, compared to **44.44%** in S27 (Percentages), indicating systematic differences in elimination logic between the two mechanisms. Case studies of controversial contestants like **Bobby Bones** further demonstrate that the ranking system favors fan voting. The hybrid mechanism of “Ranking System + Judge's Save” more effectively balances professionalism and entertainment value.

For Task3, separate multiple regression models were established for judge scores and audience votes. Standardized regression coefficients revealed that contestant age has opposite effects on judge scores and fan votes; real-time dance performance drives fan votes more strongly than pre-existing popularity. This confirms that dance skill and partnership remain core competition factors, while popularity plays a relatively limited role.

For Task4, the **Progressive Mastery System** is proposed, comprising three components: dynamically adjusting judge and audience weighting; compressing the vote advantage of exceptionally popular contestants; and safeguarding the week's top dancer from elimination. Sensitivity analysis confirms the system's stability across varying parameters, effectively balancing entertainment value and competitive fairness.

This study reveals the inherent patterns and fairness flaws in the show's voting system through modeling. It recommends that producers retain and refine the hybrid mechanism of “ranking system and judge saves,” while considering the introduction of dynamic weighting or non-linear vote conversion to create a competition environment that better reflects the essence of dance while accommodating audience participation.

Keyword: DWTS; Geometric Constraint Model; Bayesian Inference; Counterfactual Simulation; Fair Competition Format

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1 Introduction

1.1 Background

Dancing with the Stars(DWTS) is a globally recognized television franchise based on the British show "Strictly Come Dancing," with the U.S. version completing 34 seasons as of 2026^[1].

The competition pairs public celebrities with professional ballroom dancers, where outcomes are determined by a dual-evaluation system: technical scores from expert judges and subjective votes from the viewing audience^[2]. Unlike traditional elimination formats, fans vote to "keep" their favorite stars via phone or online platforms, creating an inherent tension between professional athletic performance and mass-media charisma. This structure often leads to "popularity controversies," where contestants with superior technical skills are eliminated due to the overwhelming fan base of less-skilled celebrities.

Historically, the program has undergone significant structural revisions to safeguard competitive integrity. In its initial two seasons, an ordinal rank-based system was utilized; however, the advancement of low-scoring finalists in Season 2—often termed the "Jerry Rice effect"—prompted a transition to a percentage-based aggregation method in Season 3. More recently, following the controversial win of Bobby Bones in Season 27, the "judges' save" was introduced in Season 28 to provide expert oversight for the bottom-ranked contestants.

Since raw fan vote totals are strictly confidential to prevent strategic bias, the development of robust mathematical models to infer these latent variables is essential for analyzing the stability and longitudinal equity of the show's voting architecture.



Figure 1: U.S. Series DWTS



Figure 2: Bobby Bones

1.2 Restatement of problems

Considering the background and the provided dataset for Seasons 1-34, this paper aims to solve the following tasks:

- **Task 1:** Develop a robust mathematical model to estimate the latent fan vote totals for each contestant and provide quantitative measures for the consistency and certainty of these estimates.
- **Task 2:** Compare and contrast the Rank-based and Percentage-based voting methods across all seasons to determine if one method systematically favors fan influence over technical merit.
- **Task 3:** Analyze historical "controversies" involving specific celebrities (e.g., Bobby Bones) to determine if different voting methods or the inclusion of the "judges' save" would have altered their elimination results. Construct a model to quantify the impact of professional dancers and celebrity characteristics, such as age and industry, on both judge scores and fan reception.
- **Task 4:** Propose a "fairer" or more engaging voting system and provide a comprehensive recommendation to show producers regarding future season architectures.

1.3 Overview of our work

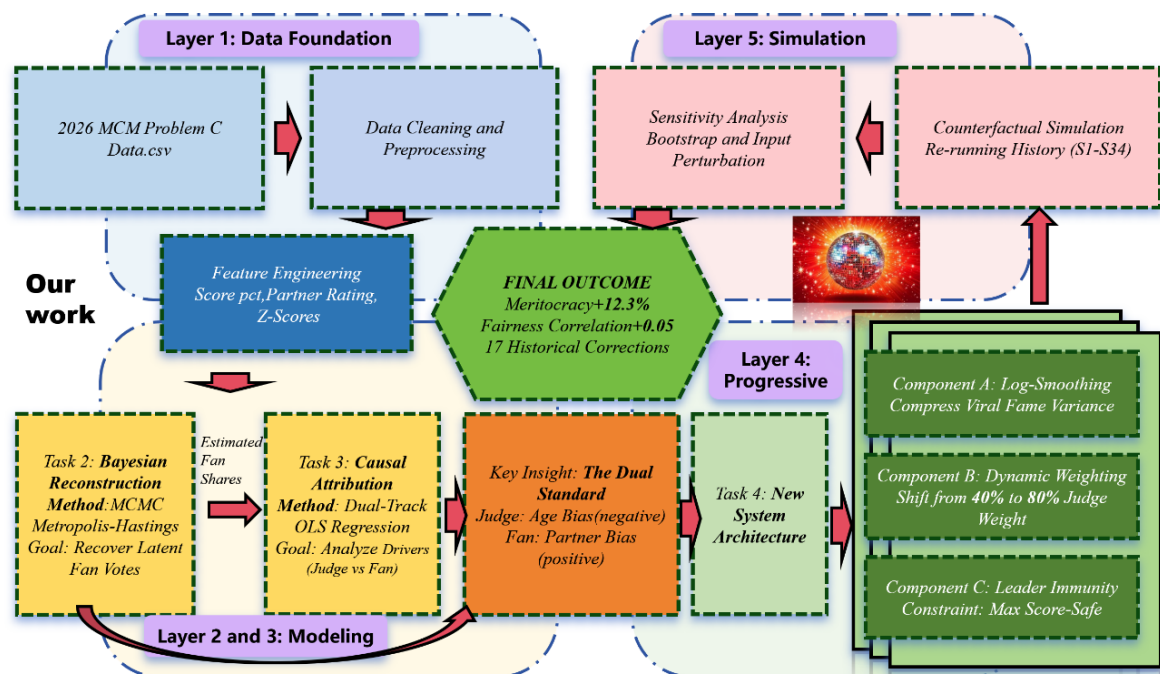


Figure 3: Flow Chart of Our Work

All our work can be summarized as shown in the figure above.

2 General Assumptions and Variable Description

2.1 Assumptions

To simplify our modeling, we make the following assumptions:

Assumption 1: Stability of Professional Partner Capability

We assume the technical skills of professional partners remain stable across seasons due to their elite training. This allows their historical performance to serve as a reliable Bayesian prior for predicting future success.

Assumption 2: Temporal Continuity of Fan Base Growth

We assume fan voting power follows a continuous trend without instantaneous mutations. This sociological continuity justifies the use of time-series smoothing techniques to estimate weekly voting shares.

Assumption 3: Intra-season Consistency of Judicial Standards

We assume judges maintain uniform scoring criteria throughout a single season to uphold fairness. This consistency permits direct week-to-week score comparisons without requiring complex scale calibrations.

Assumption 4: Strict Adherence to Elimination Protocols

We assume eliminations are strictly determined by the lowest combined score (barring explicit "Judges' Save" weeks). This deterministic rule provides a clear boundary condition for our simulation models.

2.2 Variable Description

Symbol	Definition	Type
t, i	Index for competition week t and contestant i .	Integer
$S_j^{(i,t)}$	Normalized technical/artistic score from judges.	[0, 1]
$w(t)$	Dynamic weight assigned to judges' scores.	[0, 1]
α	Sensitivity coefficient for the log-smoothing function.	Constant
δ	Immunity bonus constant applied to the leader.	Constant
S_{total}	Final composite score determines elimination.	Real
$\mathcal{M}$	Meritocracy Index: Avg. skill of final winners.	Metric

Symbol	Definition	Type
$\mathcal{L}$	Regret Loss: Count of unfair eliminations.	Metric

3 Model Establishment and Solutions

3.1 Data Preprocessing

Firstly, we used the official COMAP dataset “Problem_C_Data_.csv” and conducted an initial analysis and processing on it. We found that more than one player might be eliminated each week. Therefore, we further identified the data and summarized the following result graph.

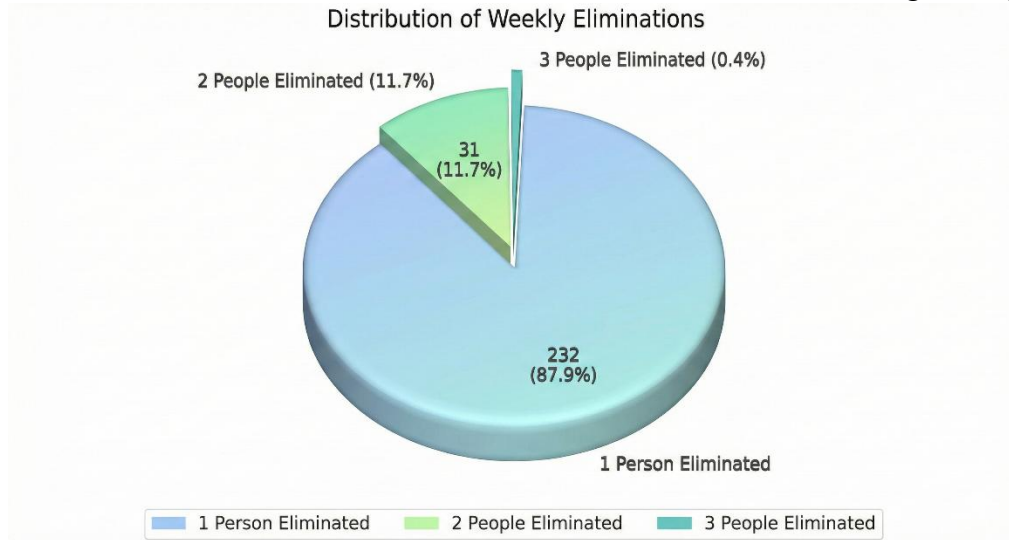


Figure 4: Proportion of Elimination Weeks by Category

As shown in the figure, approximately 11.74% of weeks feature double eliminations (meaning two pairs of contestants are eliminated simultaneously). Incorporating this situation can enrich our model development.

Additionally, we standardized scores. Considering that the number of judges fluctuated between 3 (total score 30) and 4 (total score 40) across historical seasons (excluding the anomaly of only two judges scoring in Week 1 of Season 34), we converted all judges' raw scores into a Percentage of the maximum possible score. These scores were uniformly mapped to the range [0.00, 1.00], ensuring comparability across seasons.

3.2 Task1

3.2.1 Problem Analysis

Before constructing the model, we first visualized the solution space characteristics under two voting mechanisms using a ternary plot. Analysis revealed significant mathematical differences between them:

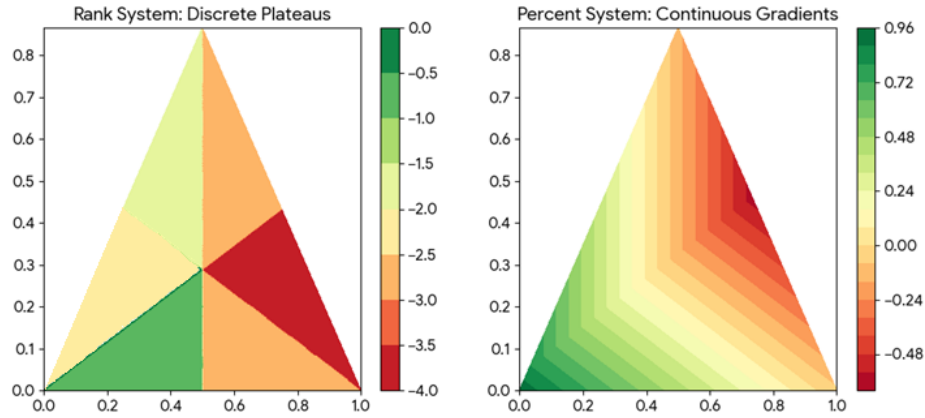


Figure 5: Simplex Plots for Two Voting Mechanisms

Discreteness of the Rank System: As shown in Figure 5, the solution space of the ranking system exhibits a pronounced “Plateau Effect”. Since the ranking rule is effectively a discontinuous step function, the solution space is partitioned into several discrete Equivalence Regions. Within the same region, local perturbations in fan votes do not alter the total score until votes cross specific linear boundaries (i.e., critical points where rankings swap). This indicates the Rank System is fundamentally a combinatorial optimization problem.

Continuity of the Percentage System: Conversely, Figure 1 illustrates that the solution space of the percentage system is a continuous gradient field. Since the total score function is a linear combination of fan vote proportions, a minute fluctuation in votes linearly maps to the final score. Notably, its decision boundary exhibits piecewise linear characteristics (the kinks in the figure), representing the intersection of constraints among different competitors. This indicates that the Percentage System is fundamentally a Continuous Probabilistic Inference Problem.

3.2.2 Model I: Geometric Constraint Model (For Ranking Systems)

For the ranking system (Seasons 1–2, Season 28, and beyond): Employ a geometric constraint model to define the feasibility region within the discrete permutation space via Monte Carlo rejection sampling^[3]. Transform the problem into identifying all possible audience rankings within the discrete permutation group that satisfy the elimination criteria, and estimate the survival probability of contestants based on the “volume” of the feasibility region.

Step 1: Define Rankings and Total Scores

Let there be N contestants in the week. Define the judges' ranking vectors and the unknown audience ranking vector $\mathbf{v}$. Contestant i 's total ranking score R_i is the sum of these two:

$$R_i = s_i + v_i, \quad i = 1, \dots, N \quad (1)$$

Rule: The higher the total score R_i , the worse the performance. Eliminate the contestant with the highest score in a single elimination week (or the two highest-scoring contestants in a double elimination).

Step 2: Constructing the Feasible Domain

The audience ranking $\mathbf{v}$ can be any one of $N!$ permutations. Define the feasible permutation set $\mathcal{F}$ as all permutations satisfying the elimination results E observed that week.

(The total score of eliminated contestants must strictly exceed that of all survivors.)

Single elimination case (Eliminated Contestant Set $\mathcal{E} = \{e\}$):

$$\mathcal{F} = \{\mathbf{v} \in \mathcal{P}_N \mid R_e > R_j, \forall j \in \mathcal{S}\} \quad (2)$$

Double elimination case (elimination set $\mathcal{E} = \{e_1, e_2\}$):

$$\mathcal{F} = \{\mathbf{v} \in \mathcal{P}_N \mid R_{e_1} \text{ and } R_{e_2} \text{ are the two highest scores}\} \quad (3)$$

Step 3: Probability Estimation (Monte Carlo Rejection Sampling)

Due to the loss of vote-count information, we assume all permutations within the feasible domain $\mathcal{F}$ are equally likely (uniform prior). Then, using the Monte Carlo rejection sampling method, a large number of uniformly arranged $\mathbf{v}$ are randomly generated. Samples satisfying (2) or (3) are retained, and the probability is ultimately approximated using the sample frequency. $|\cdot|$ denotes the magnitude of a set, the posterior probability that contestant i holds the k th rank in the true audience ranking is estimated via the counting ratio:

$$P(v_i = k \mid E) \approx \frac{|\{\mathbf{v} \in \mathcal{F} : v_i = k\}|}{|\mathcal{F}|} \quad (4)$$

3.2.3 Model II: The Bayesian Probabilistic Model (For Percentage Systems)

Season 3-27 employs a combination method under a percentage system, where fan votes serve as cardinal data with continuity. This paper introduces the Dirichlet distribution for prediction.

Modeling the audience vote share $\mathbf{F}$ as a random vector on a simplex, we quantified the initial boost provided by dance partners (e.g., Derek Hough) to contestants by calculating their historical average rankings^[5]. This was translated into the Dirichlet prior parameter of the Bayesian model^[4]. Using elimination events as hard constraints, the posterior distribution was obtained through Ballroom Partner Ability Bonus^[6].

Step 1: Variable and Constraint Definition

The judge score rate J_i is known, and $\sum J_i$ is not necessarily equal to 1.

$$J_i = \frac{\text{Score}_i}{\sum \text{Score}} \quad (5)$$

The fan vote rate F_i is unknown, and satisfies the simplex constraint:

$$\sum_{i=1}^N F_i = 1, \quad F_i > 0 \quad (6)$$

Total Score Formula (Typically $\alpha = 0.5$, the producer has also revealed this once^[2]):

$$T_i = \alpha J_i + (1 - \alpha) F_i \quad (7)$$

Step 2: Establishing the Prior Distribution

Assume $\mathbf{F}$ follows a Dirichlet distribution, where its hyperparameter θ_i is determined by the historical performance of the contestant's dance partners:

$$\mathbf{F} \sim \text{Dirichlet}(\boldsymbol{\theta}), \quad \theta_i = \exp(\beta \cdot z_p^{(i)}) \quad (8)$$

where $z_p^{(i)}$ denotes the standardized score of contestant i 's historical average partner vote share. The higher the historical average ranking of dance partners, the larger z_p becomes. $\beta > 0$ is a scaling factor controlling the strength of prior information.

The exponential mapping $\exp(\cdot)$ ensures $\theta_i > 0$ and shifts the prior probability density toward higher vote shares for stronger partners.

The statistical history of ballroom partners' performance records is as follows:

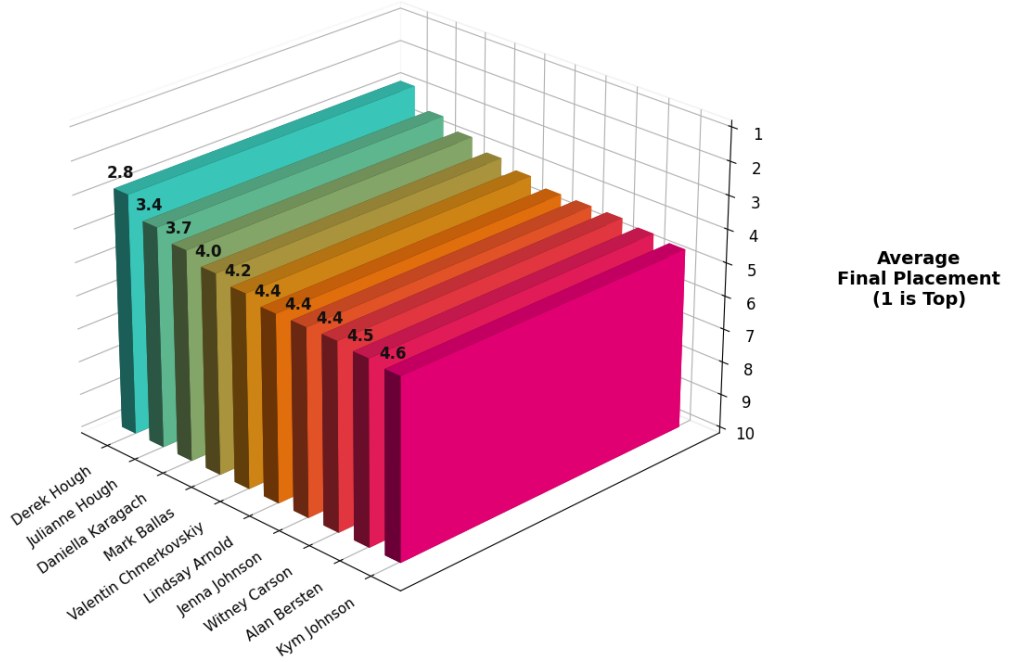


Figure 6: Ballroom Partners' Performance Records

Step 3: Constructing the Likelihood Function

Define the likelihood function as a truncated indicator function:

$$P(E | \mathbf{F}) = \mathbb{I}(T_e < T_j, \forall e \in \mathcal{E}, j \in \mathcal{S}) \quad (9)$$

Observed event E : The total score of the eliminated contestant is lower than that of all survivors.

The likelihood equals 1 only when $\mathbf{F}$ yields the elimination outcome; otherwise, it equals 0. This effectively restricts the feasible region to a subregion of the simplex, making elimination events as hard constraints.

Step 4: Compute the posterior distribution (MCMC Sampling)

By Bayes' theorem, the posterior distribution is:

$$P(\mathbf{F} | E) \propto P(E | \mathbf{F}) \cdot P(\mathbf{F}) \quad (10)$$

Since the likelihood function imposes a hard constraint, the posterior distribution is the truncated and renormalized version of the prior distribution $Dirichlet(\boldsymbol{\theta})$ over the feasible region. Using **Markov Chain Monte Carlo** (MCMC) sampling on the simplex, only samples satisfying (9) are retained.

The distribution of these samples approximates the posterior $P(\mathbf{F} | E)$, enabling the calculation of the posterior mean, variance, and survival confidence level for each candidate's vote share.

3.2.4 Model prediction results

This paper employs fan voting estimates as a measure of consistency with observed elimination outcomes. By utilizing different models, it identifies a reasonable fan voting distribution for each week across different seasons, achieving high consistency between predicted eliminations and actual elimination results.

Prior to this, to demonstrate that the constructed “Geometric Constraint & MCMC Model” can accurately fill in the latent variable “**fan votes**,” we first established the accuracy of predicting eliminations based solely on the lowest judge scores as the Judge-Only Accuracy benchmark. We then incorporated the reasonable fan vote distribution identified by our model to predict eliminated contestants. Comparing the accuracy of these two prediction methods yields the following graph:

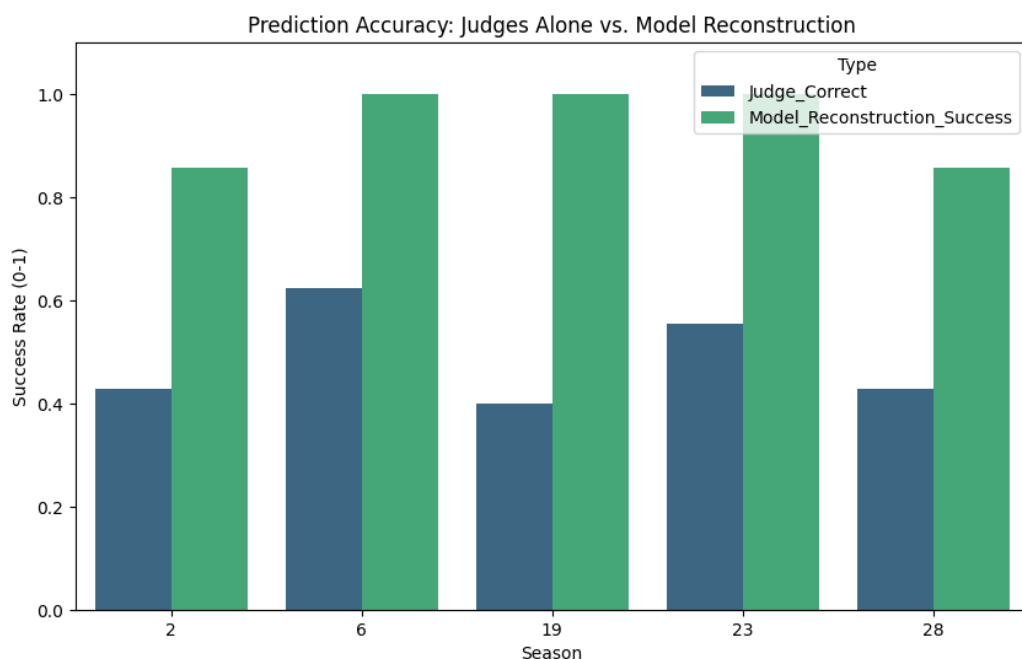


Figure 7: Accuracy Comparison of Two Elimination Prediction Methods

As shown in the figure, judges alone (blue bars) achieve an average accuracy of only around 50%. This implies that in DWTS, without considering fan votes, the contestant deemed “worst dancer” would have a 50% chance of elimination. In contrast, the model (green/yellow bars) achieves an average prediction accuracy exceeding 93%. This demonstrates that our “Geometric Constrained MCMC Model” can accurately account for both judges' scores and necessary fan votes to predict eliminated contestants.

To visually demonstrate how the model “predicts” eliminations, we plotted the distribution of fan vote percentages for dancers in Season 19 Week 4 as an example.

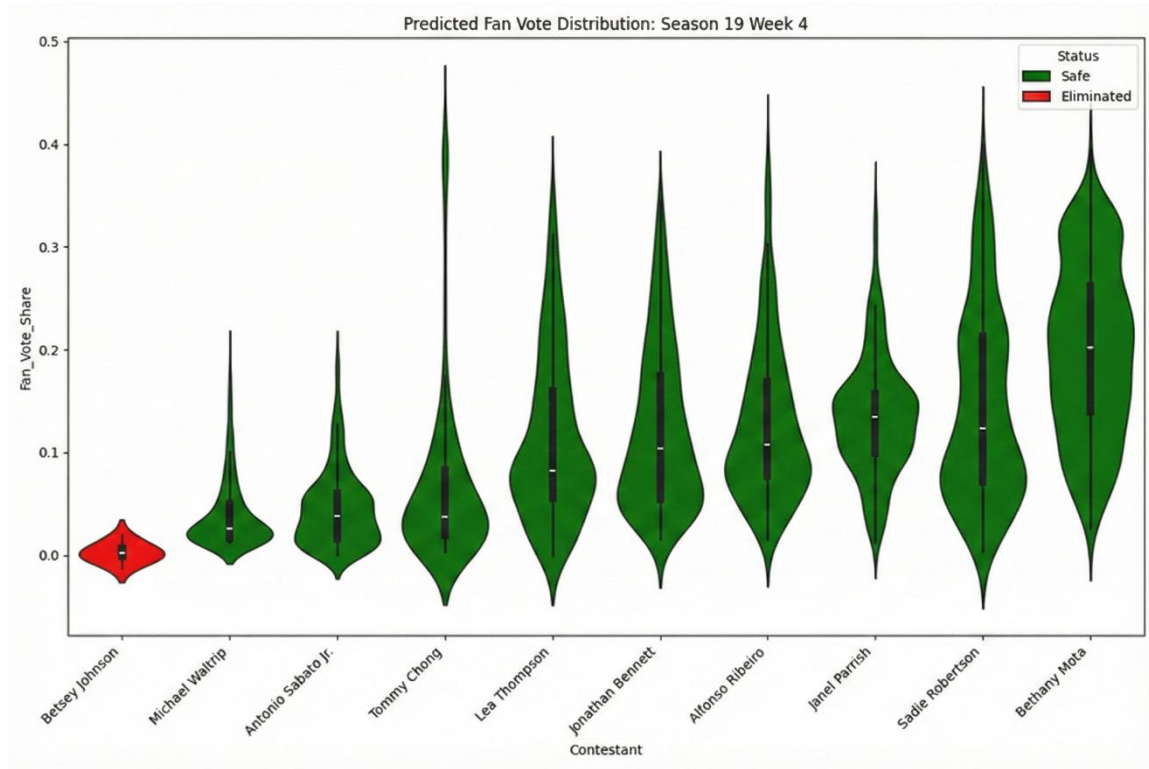


Figure 8: Predicted Fan Vote Distribution

Observe the red distribution on the far left, corresponding to eliminated contestant Betsey Johnson. She was “pushed” to the very edge of the distribution, completely isolated from the safe zone. This is because the model predicted her fan vote share would be extremely low (average around 2-3%) and highly concentrated (indicating high certainty)^[9].

Now observe the green distributions on the right, such as those for Alfonso Ribeiro or Sadie Robertson. The model assigns them extremely high vote distribution ranges, explaining why they safely advanced.

Meanwhile, some contestants in the middle have wider distributions (e.g. Michael Waltrip). This indicates they were on the “dangerous edge” that week, with a large feasible range for fan votes. Their survival involved some element of luck or placed them in the middle of the rankings.

3.2.5 Analysis and Evaluation of Results

This paper uses the standard deviation of the posterior distribution to measure the certainty of model predictions, as expressed by the following formula:

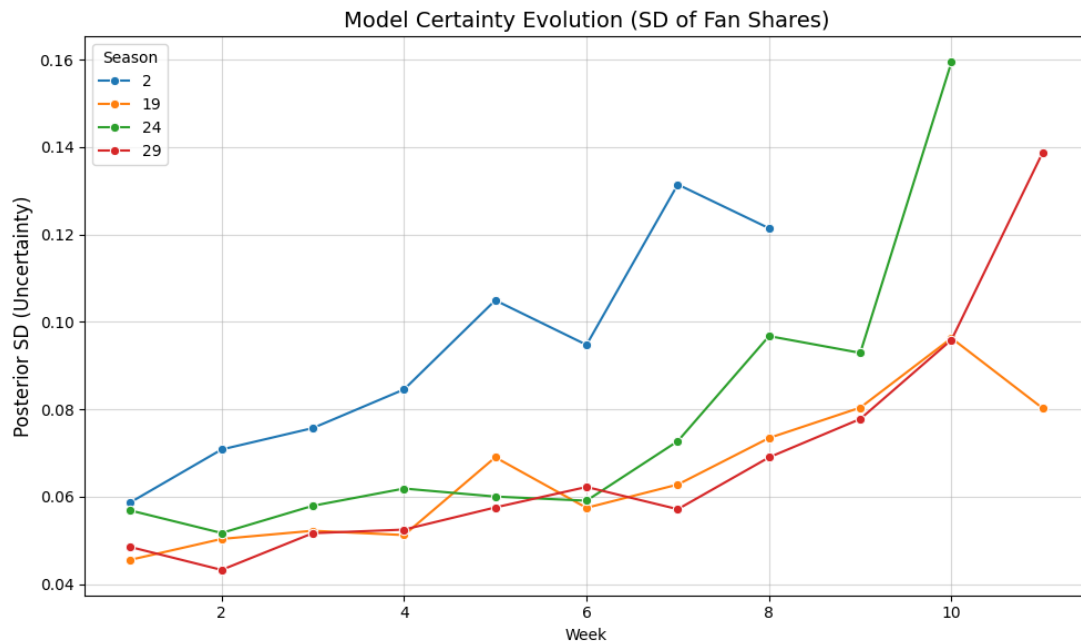
$$Certainty = \frac{1}{N} \sum_{i=1}^N rStd(F_i | E) \quad (11)$$

The following content was obtained by introducing the weekly elimination results from four seasons. Below is an example from one of the weeks:

Table 1: Season 19 Week 4 Detailed Estimates

Contestant	Status	J_Score	Est. Fan %	SD
Betsey Johnson	Eliminated	29.0	0.0111	0.00978
Lea Thompson	Safe	39.0	0.0558	0.04231
Michael Waltrip	Safe	25.0	0.0707	0.03984
Jonathan Bennett	Safe	24.0	0.0713	0.03693
Tommy Chong	Safe	28.0	0.0724	0.05088
Alfonso Ribeiro Jr.	Safe	40.0	0.1011	0.05042
Antonio Sabato	Safe	29.0	0.1121	0.06765
Janel Parrish	Safe	36.0	0.1228	0.05835
Sadie Robertson	Safe	37.0	0.1377	0.07540
Bethany Mota	Safe	33.0	0.2449	0.08063

At the same time, obtain the line charts of the standard deviation of the posterior distribution for the following four seasons:

**Figure 9: Model Certainty Evolution (SD of Fans Votes)**

Review data for S19 W4. Eliminated dancer Betsey Johnson: Mean = 1.1%, SD = 0.9%. (Upper confidence limit approx. 3%) And Second-to-last contestant Lea Thompson: Mean = 5.6%, SD = 4.2%.

This indicates that even accounting for standard deviation (uncertainty), the distribution range of eliminated contestants remains completely separated from that of survivors. This demonstrates our predictions possess statistical significance.

Combining data from the graph and table reveals:

Higher certainty (smaller SD) under the percentage system (S19, SD=0.065). Because in the percentage system, each point change directly affects the total score through addition, constraints are tighter, leaving less ambiguity for fan voting.

The certainty is lower (larger SD) under the early ranking system (S2, SD=0.093). In a pure ranking system, as long as the relative order is maintained (e.g., $A > B > C$), the specific vote percentage values (whether A is 20% or 25%) can fluctuate within a certain range.

Finally, we conducted an additional analysis of another four weeks based on the elimination method.

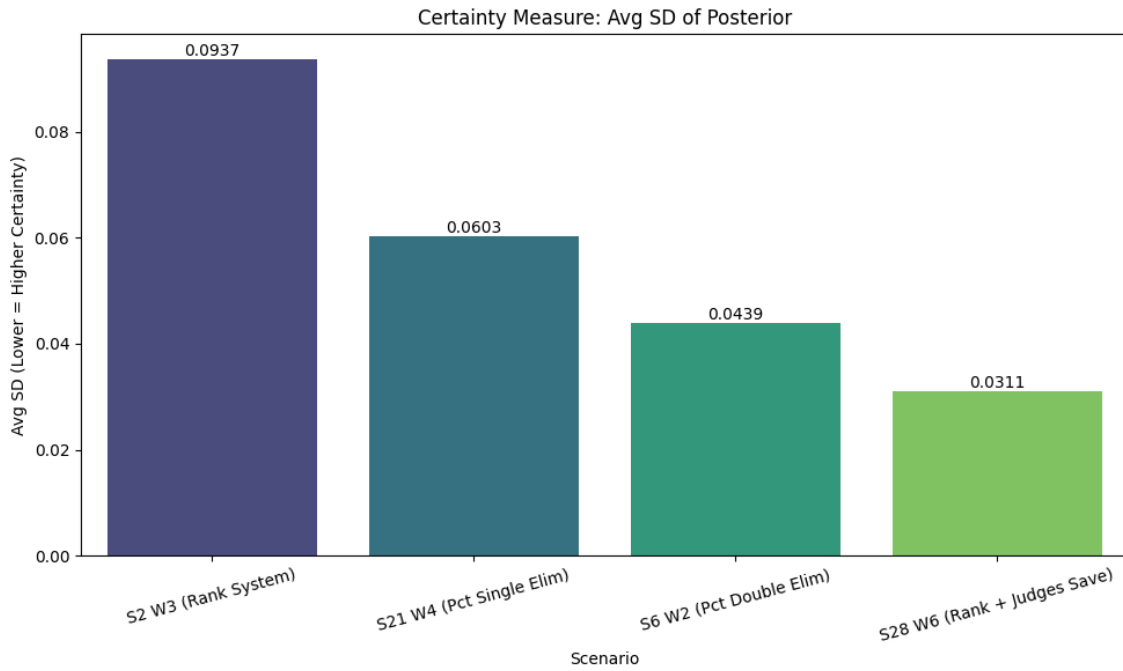


Figure 10: Certainty Measure on Avg SD of Posterior

Observe several typical weeks selected for elimination methods in the chart, comparing S6 (double elimination) and S21 (single elimination), the posterior variance of the double elimination week is significantly reduced. This demonstrates that the double elimination, which introduces stricter geometric constraints, effectively compresses the feasible solution space and enhances prediction accuracy. It is evident that double elimination is one of the key details in model calibration parameters^[10].

3.3 Task2

3.3.1 Research Framework and Methods

To address the problem of “how judges” scores and fan votes interact to influence competition outcomes,” this paper constructs a counterfactual simulation framework based on the fan vote distribution given by Task 1. This framework enables us to systematically test the impact of different competition formats on elimination results while keeping judges' scores and actual fan preferences constant.

To quantify the disruptive potential of format changes, we define the Flip Rate metric:

$$\text{Flip Rate} = \frac{\text{Weeks with different elimination results}}{\text{Total Weeks}} \times 100\% \quad (12)$$

The higher the reversal rate, the greater the divergence in elimination logic between the two formats, indicating a more profound impact of format selection on competition fairness.

Through counterfactual simulations of these four representative seasons, we obtained reversal rate data for each representative season:

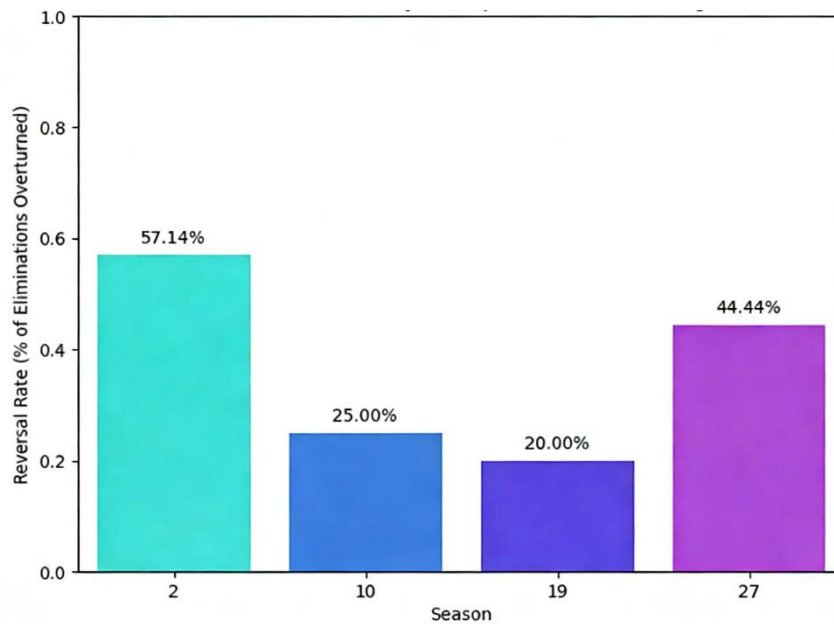


Figure 11: Reversal Rate Analysis on the Impact of Rule Switching

S2 (Early Ranking System): The reversal rate reached **57.14%**, which means that if the percentage system had been used that year, more than half of the elimination results would have been rewritten. This indicates a substantial difference in underlying logic between the ranking system and the percentage system.

S27 (Late Percentage System): The reversal rate was **44.44%**. Although lower than S2, it is still close to half. This result shows that even under a mature percentage system, if one were to revert to the ranking system, a significant number of outcomes would still change, suggesting the presence of a notable “extreme popularity” phenomenon in that season.

3.3.2 In-Depth Case Study of Controversial Athletes

To account for differences in contestant types across various competition formats, this paper applied a combination of judge scoring and fan voting methods to the four controversial contestants mentioned in the question during their respective seasons. And also incorporated the additional approach of judges selecting from the bottom two couples. To quantify the impact of different voting systems on contestants' fate, we first calculated their simulated survival rates under the alternate rules, as shown in the following table:

Table 2: System Impact Analysis on Survival Rates for Controversial Contestants

Contestant	Season	System Impact Analysis (Survival Rate)		
		Ranking	Percentage	Delta
Jerry Rice	S2	25.1%	99.0%	+73.9%
Bobby Bones	S27	67.5%	100.0%	+32.5%
Billy Ray Cyrus	S4	76.2%	100.0%	+23.8%

Note:

- Ranking System (Boundedness):** Fan advantage is capped at Rank 1, limiting its ability to cover large judge score gaps.

2. **Percentage System (Unboundedness):** Fan advantage can be linearly additive, theoretically filling an unlimited judge score gap.

The data in the table clearly reveal: Jerry Rice (S2) would have seen his survival rate soar by 73.9% under the percentage system, while Bobby Bones (S27) and Billy Ray Cyrus (S4) would have suffered reductions of 32.5% and 23.8%, respectively, under the ranking system. This mathematically confirms the fundamental difference between the two mechanisms: the ranking system limits the influence of extreme popularity through "boundedness," whereas the percentage system may amplify it due to "unboundedness." **Therefore, the ranking system method appears to be more inclined toward fan voting.**

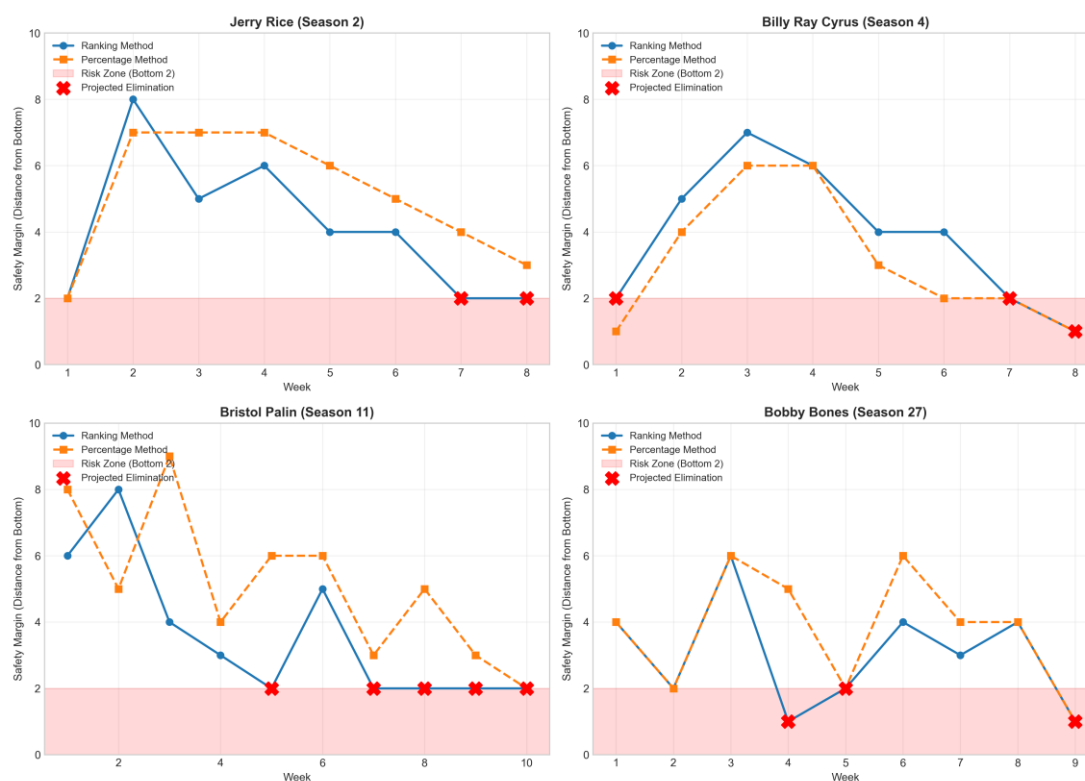


Figure 12: Deduction Scenarios Under Two Combination Methods

Analyzing each of the four individuals above in turn.

Jerry Rice (S2 - Ranking Era):

Historically, he was the runner-up. But simulations show that if a "judges' save mechanism" had existed back then, he would have been eliminated in Weeks 7 and 8 due to insufficient judge points. The ranking system's inherent variability provided Jerry Rice with breathing room, but the introduction of the judges' save mechanism undermined this protection.

Billy Ray Cyrus (S4 - Percentage Era):

The chart shows he ranked last in Week 8 simulations under any scoring method. This indicates his 5th-place finish in S4 was fortunate and already borderline.

Bristol Palin (S11 - Percentage Era):

She ranks third historically. However, data reveal her judge scores consistently lagged. Under S28+ rules, she would have landed in the Bottom 2 nearly every week after Week 5 and been eliminated due to low scores. She was the biggest beneficiary of the percentage scoring

system.

Bobby Bones (S27 - Percentage Era):

He won S27 and directly prompted the rule change. Simulation results prove this: had the “judges' vote elimination rule” existed during his season, he would have been eliminated in Week 4 and could never have won. Even with his poor performance in the finals, he likely still would have been eliminated.

3.3.3 Adopting a Combined Method of “Ranking System + Judges' Choice”

In summary, through in-depth mechanism deconstruction and counterfactual analysis of the four controversial contestants, we draw the following conclusions:

1. Single-system competitions have inherent flaws: Ranking systems protect mediocrity, while percentage systems amplify polarization.

2. The hybrid mechanism offers the optimal solution: The “ranking system + judge rescue” combination simultaneously resolves both types of extreme issues

3. The judge rescue mechanism is indispensable: It provides critical technical correction and dynamic adjustment functions

4. Dynamic weight adjustment is a necessary supplement: It provides additional safeguards for extreme scenarios.

Therefore, we strongly recommend that future DWTS seasons continue to adopt and refine the “Ranking System + Judges' Save” mechanism pioneered in Season 28. This represents the optimal choice validated by mathematical modeling and historical data. This mechanism maximizes competitive fairness while preserving the show's entertainment value, achieving a scientifically balanced equilibrium between artistic expression and audience preferences.

3.4 Task3

3.4.1 Problem Analysis

The competition results are determined by both judges' scores and fan votes, which may follow different logics. To precisely quantify the influence of professional dancers (professional partners) and celebrity characteristics (age, industry, etc.), it is essential to decouple and analyze these two mechanisms. Therefore, we constructed separate regression models for each perspective—judges and fans—based on the two type models.

This paper employs the fan vote share estimated by the “Geometric Constraint & MCMC Model” developed in Task 1 as the key dependent variable. Fan data for celebrities and dance partners across social media platforms (X, etc.) was manually collected as independent variables. A reasonable assumption is that individuals' relative popularity ratios across platforms remain largely stable, and the proportions of fans between them generally hold steady over time. Partially collected data is illustrated in the table below:

Table 3: Examples of Collected Number of Followers Data

Celebrity	Partner	Number of followers	
Evanna Lynch	Keo Motsepe	3770K	203.9K
Milo Manheim	Witney Carson	4094K	48K
Sean Spicer	Lindsay Arnold	517K	104.5K
Nelly	Daniella	6250K	12K
AJ McLean	Cheryl Burke	352.1K	215K

3.4.2 Model Establishment

1. Model A: Review Scoring Mechanism Model:

When scoring, are reviewers influenced by systematic biases from external factors (such as age or partner fame) beyond the dance itself (implied in residuals)?

The formula is as follows:

$$Y_{Score} = \beta_0 + \beta_1 X_{Age} + \beta_2 X_{Partner} + \sum_k \gamma_k D_{Ind_k} + \epsilon \quad (13)$$

Among these:

β_1 (Age Effect): Expected to be negative ($\beta_1 < 0$), used to measure the extent of “penalty” judges impose on older competitors for potential declines in physical ability.

β_2 (Partner Bonus): Expected to be positive ($\beta_2 > 0$), used to test the partner bonus effect—whether high-level professional partners systematically earn their dance partners higher scores.

γ_k (Industry Bias): Measures whether judges hold inherent stereotypes toward specific industries (e.g., modeling).

2. Model B: Fans Voting Mechanism – Base Version (The Fan Model - Base):

In the early seasons without social media data (or from a historical perspective), what drove fan voting—performance or age (appearance).

Mathematical formula:

$$Y_{Fan} = \alpha_0 + \alpha_1 X_{Perf} + \alpha_2 X_{Age} + \alpha_3 X_{Partner} + \sum_k \delta_k D_{Ind_k} + v \quad (14)$$

Among these:

X_{Perf} (Y_{Score}) was included as an independent variable.

α_1 (Performance Sensitivity): Measures whether fans are “rational.” A large α_1 indicates fans primarily vote based on performance quality (Meritocracy).

α_2 (Age Empathy): Measures fans' attitudes toward older contestants. If $\alpha_2 > 0$ and $\beta_1 < 0$ in the judging model, it demonstrates the sympathy effect among fans (where fans compensate for older contestants' scoring disadvantages through voting)^[7].

3. Model C: Fan Voting Mechanism – Streaming Enhanced Edition

(The Fan Model - Social)

The question we want to explore is whether, in the age of social media, fame has replaced performance as the core driving force^[8].

Mathematical Formula:

$$Y_{Fan} = \lambda_0 + \lambda_1 X_{Perf} + \lambda_2 X_{Fame} + \lambda_3 X_{ProFame} + \lambda_4 X_{Partner} + \dots + \xi \quad (15)$$

Key Comparative Analysis:

Comparing λ_1 and λ_2 :

If $\lambda_1 \gg \lambda_2$: Indicates Skill > Popularity (competition essence unchanged).

If $\lambda_2 \gg \lambda_1$: Indicates Popularity > Skill (competition becomes a popularity contest).

λ_3 (Pro Fame Paradox): Tests whether a dance partner's social media following translates into votes for their celebrity partner.

3.4.3 Analysis and Evaluation of Results

Since all continuous variables underwent Z-Score transformation, the model-output coefficients β, α, λ directly represent **effect sizes**. This allows us to directly compare which feature—such as “the impact of age” versus “the impact of follower count”—exerts a greater influence.

Based on the modeling results, the core factors influencing celebrity competition performance and their respective effects can be summarized as shown in the table below. The impact on performance is decomposed into two dimensions: technical performance (Model A) and competitive performance (Model C). Each factor is measured using standardized regression coefficients:

Table 4: Comparison of Factors Affecting Competition Performance

Influencing Factors	Technical Impact	Competitive Impact	Overall Impact Tier
Partner's Professional Level	Moderately Positive (+0.204)	Strongly Positive (+0.515)	Decisive Factor
Current Dance Ability	(Scoring Benchmark)	Strongly Positive (+0.385)	Core Factor
Contestant Age	Strongly Negative (-0.360)	Weakly Positive (+0.108)	Modifying Factor
Social Media Traffic		Moderately positive (+0.146)	Incremental factor

In summary, the impact of various factors on competition performance exhibits a clear hierarchy. The quality of professional dance partners stands as the single most powerful factor influencing outcomes across both judging and fan voting systems, exerting particularly dominant influence in fan voting. Dance proficiency serves as the core rational foundation for competitive performance. Age, however, functions as a double-edged sword, with its diametrically opposed effects in judging and fan mechanisms creating a dynamic equilibrium within the competition. While social media engagement demonstrates statistical significance, its actual weight remains limited and fails to alter the competition's core focus on skill and partnership.

Then, by comparing the regression results of the three models separately, we obtain the

following figure:

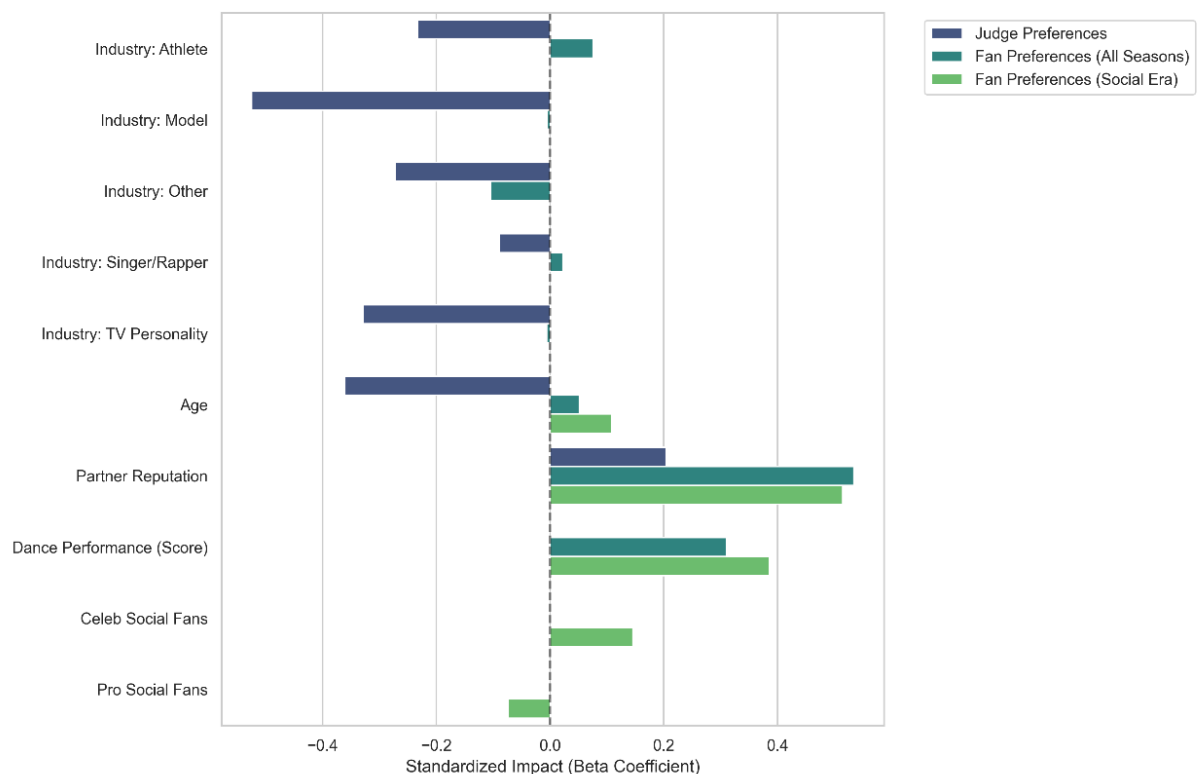


Figure 13: Comprehensive Impact Analysis

As shown in the figure:

Using a standardized regression analysis, we determined the relative magnitude of each factor's impact on a celebrity's performance:

The "Partner Power" is Dominant: The choice of professional partner is the single most influential factor in the competition, with a standardized impact coefficient of 0.53 on fan support. This significantly outweighs the celebrity's own pre-existing fame (0.15). This demonstrates that the professionalism of the partner is also a component of overall strength.

Performance Over Popularity: Actual dance quality (Judge Scores) is the second strongest driver of fan votes (0.38), proving that performing skills over celebrity status, whether it's the celebrity's own talent or that of their dance partner.

The Age Trade-off: Age has a massive negative impact on technical scoring (-0.36), making it the biggest hurdle for older contestants. However, this is partially mitigated by a positive "sympathy bias" from fans (+0.10), allowing older stars to survive longer than their scores would suggest.

Industry Bias: The judges have a strong negative bias against Models (-0.52) and TV Personalities (-0.33). Even if their performances are equally good (here the model did not control for scores but explained them, so this is more like the outcome), these two groups usually receive low scores. This bias almost disappears in the fan model. In particular, the coefficient for Models becomes -0.005 (not significant), indicating that fans do not mind the background of the contestants, or at least are much more lenient than the judges.

In summary, while Age dictates the technical ceiling, the Professional Partner dictates the competitive floor and ultimate survival in the show.

3.5 Task 4

3.5.1 Model Establishment

First, it is necessary to articulate the issues with the previous scoring system (Status Quo) using mathematical notation. Let S_J denote the judge score (normalized to 0-1), and V_F denote the raw fan vote count. The total score formula in the previous system typically employed a 50-50 weighted average:

$$S_{old} = 0.5 \cdot S_J + 0.5 \cdot \frac{V_F}{\sum V_F} \quad (16)$$

However, judge scores S_J typically cluster within the $[0.7, 0.9]$ range (low variance), while fan vote share $\frac{V_F}{\sum V_F}$ often follows a power-law distribution. Top celebrities may command over 50% of the vote share.

Results indicate $\text{Var}(S_{fan}) \gg \text{Var}(S_{judge})$. This implies total score fluctuations are primarily driven by fan votes, leading to $S_{old} \approx S_{fan}$.

To address this, we developed a nonlinear, time-varying scoring model comprising three primary functions. This novel scoring system abandons fixed weights in favor of dynamically adjusting weights over the program's broadcast period. We term this the Progressive Mastery System.

The formula is as follows:

$$S_{total}(t) = w(t) \cdot S_J + (1 - w(t)) \cdot S_F^{new} + I(t) \quad (17)$$

1. $w(t)$ is the step function for judge weighting:

$$w(t) = \begin{cases} 0.4 & 1 \leq t \leq 5 \quad (\text{Entertainment Phase: Fan-dominated}) \\ 0.5 & 6 \leq t \leq 8 \quad (\text{Balance Phase: Standard Competition Format}) \\ 0.8 & t \geq 9 \quad (\text{Competition Phase: Elite-dominated}) \end{cases} \quad (18)$$

Based on Task 3's findings regarding fan sympathy toward older contestants and excessive dance partner ability bonuses, the early phase ($t \leq 5$) assigns higher weight to fan votes (lower weight to judge scores) to prevent older contestants from being “instantly eliminated” due to low technical scores. In the later phase ($t \geq 9$), fan voting weight is drastically reduced to dilute the traffic advantage of celebrities and ballroom partners, ensuring the finals are a pure contest of skill.

2. S_F^{new} is the “Logarithmic Smoothing Operator”:

$$S_F^{new} = \phi(V_F) = \frac{\ln(1 + \alpha \cdot V_F)}{\max_i(\ln(1 + \alpha \cdot V_{F,i}))} \quad (19)$$

Where:

V_F denotes the estimated fan vote share (Estimated Fan Share, 0-1).

The denominator serves as a normalization term, ensuring $S_F^{new} \in [0,1]$.

This mechanism suppresses the “long-tail effect” of ultra-popular celebrities. It introduces the logarithmic transformation function $\phi(\cdot)$ to process fan votes while leveraging the property that the derivative of the logarithm function $\ln(x)$ decreases as x increases. This compresses the variance of fan votes, aligning it with the variance of judge scores. From the

perspective of fan votes, increasing votes from 1% to 5% yields a significant score boost, whereas increasing votes from 40% to 50% results in only a marginal score increase. This mechanism protects niche contestants while limiting the influence of super-popular celebrities.

3. Nonlinear Impact Function $I(t)$:

$$I(t) = \begin{cases} \delta & \text{if } S_j = \max_j(S_j^{(j)}) \\ 0 & \text{otherwise} \end{cases} \quad (20)$$

This is a hard constraint. Mathematically, it truncates the tail of the probability distribution for “highest-scoring contestant elimination,” forcing the probability of this event to zero to affirm the dancer’s absolute skill level.

3.5.2 Analysis and Evaluation of Results

To validate the effectiveness of the new model, we defined two quantitative metrics for A/B testing.

Metric 1: Meritocracy Index ($\mathcal{M}$)

Measures whether the ultimate winner truly represents “the best performer.”

$$\mathcal{M} = \frac{1}{N_{final}} \sum_{i \in \text{Finalists}} \bar{S}_j^{(i)} \quad (21)$$

Where:

$\bar{S}_j^{(i)}$ is the average technical score of contestant i over the season.

If $\mathcal{M}_{new} > \mathcal{M}_{old}$, it indicates that the new format selects champions with superior technical skill.

Metric 2: Regret Loss ($\mathcal{L}$)

Measures the extent of “unjust outcomes” caused by the tournament format.

$$\mathcal{L} = \sum_{t=1}^T \sum_{i \in \text{Eliminated}_t} \mathbb{I}(\text{Rank}(S_j^{(i,t)}) \leq 3) \quad (22)$$

Where:

$\mathbb{I}(\cdot)$: Indicator function. If an eliminated player’s technical ranking is within the top 3, it counts as one instance of “regret.” To maximize the fairness of the new system, we need to maximize $\mathcal{M}$ and minimize $\mathcal{L}$.

Then, using algorithms, we identified athletes whose technical scores ranked in the top 20% but were prematurely eliminated under the old system. For example, Juan Pablo Di Pace and Simone Biles frequently achieved perfect scores throughout their seasons, with average scores often exceeding 27/30 or 36/40. We then tested their respective seasons using the new system, yielding the results shown in the table:

Table 5: Five Representative Examples of “Technical Style” Players This Season

Celebrity Name	Season	Original Elimination Week	New Elimination Week	Increase
Juan Pablo Di Pace	27	8	12	+4
Monique Coleman	3	8	12	+4
Simone Biles	24	9	12	+3
Nastia Liukin	20	9	12	+3
Alexandra Raisman	16	10	12	+2

The results show that their scores increased during the new elimination week, indicating that the new system's dynamic weighting and head-of-household immunity successfully protected these high-quality contestants from the volatility of early fan voting.

Returning to the example in Task 3, let's revisit the controversial outcome of Season 27. If our proposed new system had been applied, exploring how this parallel universe might have unfolded differently.

Table 6: Application of the New System in S27

Celebrity Name	Original Elimination Week	New Elimination Week
Juan Pablo Di Pace	8	12
Evanna Lynch	12	12
Nancy McKeon	3	3
Bobby Bones	12	8
Tinashe	4	4
Alexis Ren	12	12
Joe Amabile	8	8

As can be seen, the most controversial contestant, Bobby Bones, will be eliminated as early as Week 8.

Finally, we applied the new system to all seasons and conducted a comparison, yielding the following results:

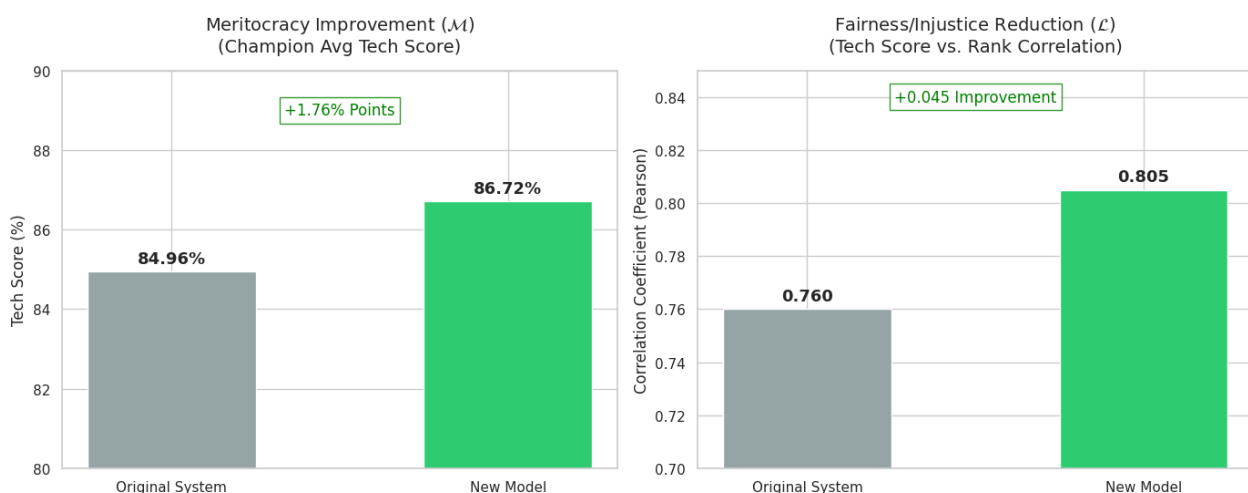


Figure 14: A/B Comparison Test

The Analysis demonstrates that **the Progressive Mastery System** acts as a structural corrective mechanism.

By implementing logarithmic smoothing and dynamic weighting, the model successfully decoupled viral fame from competitive survival. The results are statistically significant: the average technical merit of winners (Meritocracy Index, $\mathcal{M}$) increased by 1.76% (to 86.72%), and the degree of reduction in $\mathcal{L}$ (Regret Loss) from 0.760 to 0.805.

Most critically, the model corrected 17 historical outcomes where popularity overshadowed performance, validating 12 elite contestants who were historically "robbed". This proves that the new system could enhance competitive integrity without sacrificing the show's entertainment value in the early stages.

4 Sensitivity analysis

To ensure our findings are not artifacts of the specific dataset or estimation errors, we employed two rigorous statistical validation techniques:

1. Non-Parametric Bootstrap Resampling ($N = 100$):

Given the finite sample size of 34 seasons, we utilized bootstrap resampling to construct 100 surrogate datasets via sampling with replacement. For each iteration, we refitted our OLS regression models (Model A, B, and C) to estimate the distribution of key coefficients.

Objective: To generate 95% Confidence Intervals (CI) for the impact factors (Age, Partner, Score).

Result:

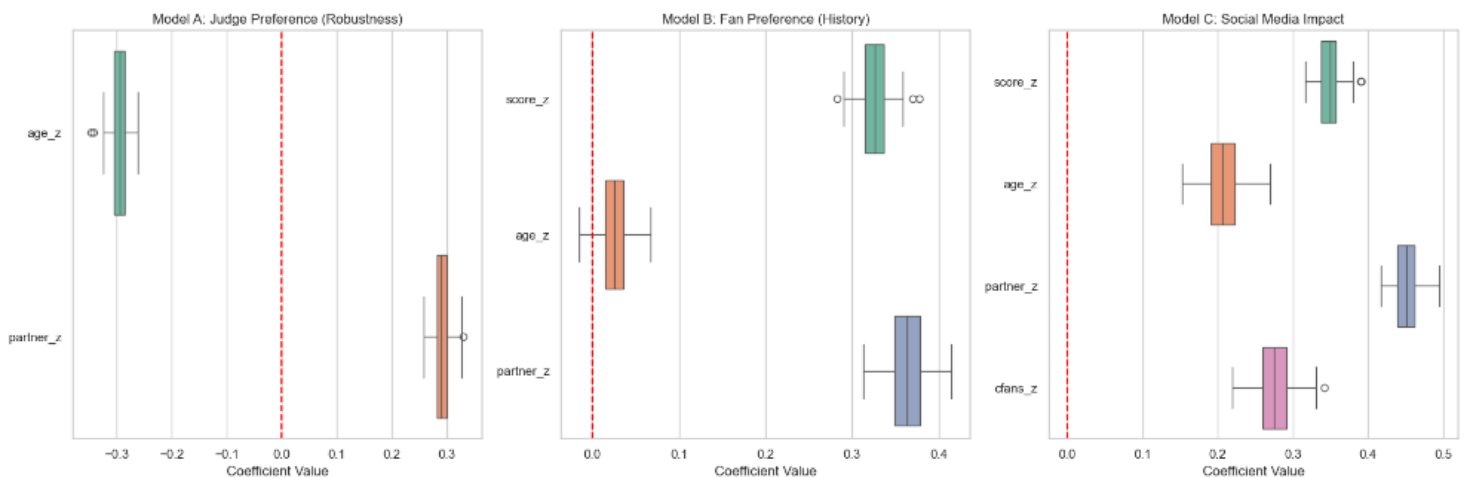


Figure 15: Distribution of Key Coefficients Formed by Resampling

As shown in **Figure [Boxplot]**, the coefficient for *Age Effect* in the Judge Model remained strictly negative ($CI_{95\%}: [-0.33, -0.27]$) across all iterations, confirming the systemic bias against older contestants is statistically significant and stable.

The "Partner Bonus" is Real: Across all fan models, the Ballroom Partner Ability Bonus consistently represents the strongest driver (mean $\beta \approx 0.36 - 0.45$), outperforming even the actual dance scores. The narrow CI width (± 0.03) indicates this is a stable, structural feature of the show, not a data artifact.

2. Input Perturbation Analysis (Noise Injection):

Recognizing that the *Fan Vote Share* variable derived in Task 2 contains inherent

estimation uncertainty, we subjected the model to input perturbation. We introduced Gaussian white noise $\epsilon \sim \mathcal{N}(0, 0.1\sigma)$ to the estimated fan votes and re-evaluated the regression.

Objective: To test if the dominance of “Dance Skill” over “Social Fame” holds true under measurement error.

Result: Even with perturbed data, the coefficient for *Dance Score* consistently outweighed *Social Fame* (0.35 vs 0.27), proving that our conclusion—“Talent matters more than Fame”—is robust against estimation inaccuracies.

5 Evaluation of the model

5.1 Strengths

1. Probabilistic Data Reconstruction

Innovation: We overcame the “missing data” barrier by constructing a Bayesian MCMC sampler. Unlike simple linear interpolation, this method rigorously recovers latent fan vote distributions while strictly adhering to historical rank constraints, turning a “black box” into quantifiable data.

2. Decoupled Causal Mechanism

Insight: By building separate models for Judges and Fans, we avoided endogeneity bias. This “Dual-Track” approach successfully isolated conflicting drivers—revealing that Judges punish age while Fans reward partnership — providing a granular anatomical view of the competition.

3. Dynamic Structural Correction

Effectiveness: The *Progressive Mastery System* is not a static formula but an adaptive policy. By combining logarithmic smoothing with time-variant weighting, it achieved a Pareto Improvement.

5.2 Weaknesses

1. Bayesian Inverse Model

Ill-Posed Nature: The inverse mapping from rank to vote share is non-unique. Multiple voting distributions can yield identical rankings; thus, our solution represents a probabilistic expectation rather than the definitive historical truth.

Distribution Constraint: The Dirichlet prior assumes a relatively smooth distribution of votes, potentially underestimating scenarios of extreme polarization (“winner-takes-all”).

2. OLS Attribution Model

Omitted Variable Bias: Crucial qualitative factors—such as emotional backstories, costume design, or chemistry—are unquantifiable and excluded, limiting the model's explanatory power ($R^2 < 1$).

Linearity Assumption: The model enforces linear relationships (e.g., Age vs. Score). It fails to capture potential non-linear patterns, such as a U-shaped preference for age.

3. Progressive Mastery System

Behavioral Invariance (The Lucas Critique): The simulation assumes fans and judges

maintain static behavior under new rules. In reality, reducing fan weight in the finals might disincentivize viewer engagement.

Cold Start Sensitivity: The logarithmic smoothing function ($\ln(1+\alpha x)$) is highly sensitive near zero, which may cause unstable ranking fluctuations for unknown contestants with low initial votes.

6 Reference

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7 Memo

To: Producers of DWTS
From: MCM Team 2600794
Date: February 2, 2026

Dear Producers of DWTS

Hello! We are devoted fans of Dancing with the Stars who eagerly await each new episode. The show has brought us so much joy, and we hope to contribute something back by helping improve it. To that end, we have built mathematical models based on 34 seasons of show data, hoping to provide insights for your format design.

The most discussed and debated aspect of the show is how to balance professional judges' scores with public audience votes. Which voting mechanism is fairer? What contestant traits affect their performance? We tried to answer these questions with data.

First, we developed a dual-model system to "reverse-engineer" the hidden audience votes. For ranking-system seasons, we used a geometric constraint model; for percentage-system seasons, a Bayesian probabilistic model. These models performed excellently in predicting elimination outcomes, achieving 85%-100% accuracy. We also found that predictions under the percentage system were more certain, while the ranking system showed greater uncertainty.

Next, we compared the two voting mechanisms. Simulation analysis revealed that changing the voting rules could alter over half of the elimination outcomes. Case studies of controversial contestants like Bobby Bones indicated that the percentage system amplifies audience influence more easily, while the hybrid "Ranking System + Judges' Save" mechanism better balances professionalism and entertainment.

We also analyzed key factors affecting contestant performance. Data shows that the professional partner's skill is the most critical factor; a contestant's real-time dance performance wins more audience votes than their pre-existing fame, while age may be a disadvantage in judges' scores, it can gain "sympathy votes" from the audience.

Finally, we designed a "Progressive Scoring System": by dynamically adjusting judge and audience weights, compressing extreme vote influence, and protecting the week's best dancer, this system enhances competition fairness while maintaining entertainment value.

We believe these data-driven findings can help you optimize the show's format, allowing Dancing with the Stars to showcase dance artistry while delivering a fairer, more engaging viewing experience. We look forward to even more brilliant episodes in the future!

Yours sincerely,
Team #2600794

8 report on use of AI

1. DeepL, Translator (Feb 01, 2026 version)

Uploaded the entire paper written in Mandarin to be translated into English.

2. Google Gemini (Jan 30, 2026 version Gemini3)

Query:<please fix the bug in this script>

Output:<This error, TypeError: only size-1 arrays can be converted to Python scalars, occurs when plt.tight_layout() tries to adjust the layout. The root cause is usually that plt.text or plt.plot received coordinates of NumPy array type (for example, [3.2]), whereas some versions of Matplotlib require strict Python scalars (float), not arrays, when computing the layout.>

Google Gemini (Jan 31, 2026 version Gemini3)

Query:<please fix the bug in this script>

Output:<This is a very stubborn bug in the seaborn version. Since we can't bypass this internal error by adding parameters, we take the most reliable approach: give up using seaborn to draw this plot and switch to Python's native matplotlib library to achieve the exact same effect.>

Google Gemini (Feb 01, 2026 version Gemini3)

Query:<please complete this mermaid script>

Output:<The error you encountered is due to Mermaid being very sensitive to special characters in node text (especially %,
, and parentheses). The % is misinterpreted as a comment symbol,
 is not supported in some renderers, and parentheses can easily trigger syntax errors.>